Extraction of α_s from Event Shapes

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Paper overview

- QCD in e^+e^- annihilations
  - General features
  - Event shape observables
- Event generation
  - Process selections
  - PYTHIA calibration
- Experimental distortions
  - QED radiation
  - Electroweak backgrounds
- **Determination of α_s**
  - **Theoretical description**
  - **Fitting procedure**
  - **Energy evolution**
- Inclusive Hadron Spectra
  - Charged hadron multiplicity
  - Momentum distributions and peaks

An Experimental Framework for QCD studies of Event Shapes and Inclusive Hadron Spectra at FCC-ee energies

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We present a comprehensive analysis of hadronic final states produced in e^+e^- annihilations at the planned FCC-ee center-of-mass energies of 91.2, 160, 240, and 365 GeV using Monte Carlo simulations with PYTHIA. The impact of enhanced electroweak distortions on event shape variables relative LEP is investigated considering initial-state photon radiation, and hadronic backgrounds arising from Z and W pairs, top-quark pairs, and Higgs decays. An extraction of the strong coupling constant α_S is performed through fits of the Thrust and C-parameter distributions to perturbative QCD predictions at next-to-next-to-leading-order (NNLO) precision, and the results are benchmarked against global values. The study further probes soft gluon dynamics through charged particle multiplicities, momentum distributions, and their energy evolution, compared with measurements from previous experiments. The learning from this phenomenological work provide a reference framework for future experimental QCD studies at high-energy electron-positron colliders.

Workflow – Theory coefficients

$(1-T)$	$(1-T)\frac{dA_{(1-T)}}{d(1-T)}$	$(1-T)\frac{dB_{(1-T)}}{d(1-T)}$	$(1-T)\frac{dC_{(1-T)}}{d(1-T)}$
0.005	$2.5797(2)\cdot 10^{1}$	$-4.957(3)\cdot 10^{2}$	$-9.0(1)\cdot 10^{3}$
0.015	$1.8299(1)\cdot 10^{1}$	$2.166(3)\cdot 10^{2}$	$-3.3(1)\cdot 10^{3}$
0.025	$1.5411(1)\cdot 10^{1}$	$2.809(3)\cdot 10^{2}$	$1.6(1)\cdot 10^{3}$
0.035	$1.35230(9)\cdot 10^{1}$	$2.920(3)\cdot 10^{2}$	$4.0(1)\cdot 10^{3}$
0.045	$1.21030(9)\cdot 10^{1}$	$2.877(2)\cdot 10^{2}$	$4.9(1)\cdot 10^{3}$
0.055	$1.09692(8)\cdot 10^{1}$	$2.772(2)\cdot 10^{2}$	$5.3(1)\cdot 10^{3}$
0.065	$1.00228(8)\cdot 10^{1}$	$2.638(2)\cdot 10^{2}$	$5.5(1)\cdot 10^{3}$
0.075	$9.2112(7)\cdot 10^{0}$	$2.505(2)\cdot 10^{2}$	$5.4(1)\cdot 10^{3}$
0.085	$8.5022(7)\cdot 10^{0}$	$2.371(2)\cdot 10^{2}$	$5.5(1)\cdot 10^{3}$
0.095	$7.8724(7)\cdot 10^{0}$	$2.240(2)\cdot 10^{2}$	$5.2(1)\cdot 10^{3}$
0.105	$7.3045(6)\cdot 10^{0}$	$2.113(2)\cdot 10^{2}$	$5.1(1)\cdot 10^{3}$
0.115	$6.7891(6)\cdot 10^{0}$	$1.993(2)\cdot 10^{2}$	$4.8(1)\cdot 10^{3}$
0.125	$6.3179(6)\cdot 10^{0}$	$1.878(2)\cdot 10^{2}$	$4.5(1)\cdot 10^{3}$
0.135	$5.8820(6)\cdot 10^{0}$	$1.768(2)\cdot 10^{2}$	$4.3(1)\cdot 10^{3}$
0.145	$5.4791(6)\cdot 10^{0}$	$1.666(2)\cdot 10^{2}$	$4.1(1)\cdot 10^{3}$
0.155	$5.1030(5)\cdot 10^{0}$	$1.572(2)\cdot 10^{2}$	$3.9(1)\cdot 10^{3}$
0.165	$4.7484(5)\cdot 10^{0}$	$1.475(2)\cdot 10^{2}$	$3.8(1)\cdot 10^{3}$
0.175	$4.4141(5)\cdot 10^{0}$	$1.393(2)\cdot 10^{2}$	$3.5(1)\cdot 10^{3}$
0.185	$4.0962(5)\cdot 10^{0}$	$1.306(2)\cdot 10^{2}$	$3.29(9)\cdot 10^{3}$
0.195	$3.7931(5)\cdot 10^{0}$	$1.231(2)\cdot 10^{2}$	$3.20(9)\cdot 10^{3}$
0.205	$3.5028(4)\cdot 10^{0}$	$1.153(2)\cdot 10^{2}$	$2.98(9)\cdot 10^{3}$
0.215	$3.2220(4)\cdot 10^{0}$	$1.081(2)\cdot 10^{2}$	$2.85(9)\cdot 10^{3}$
0.225	$2.9494(4)\cdot 10^{0}$	$1.011(2)\cdot 10^{2}$	$2.63(8)\cdot 10^{3}$
0.235	$2.6826(4)\cdot 10^{0}$	$9.47(1)\cdot 10^{1}$	$2.50(8)\cdot 10^{3}$
0.245	$2.4223(4)\cdot 10^{0}$	$8.82(1)\cdot 10^{1}$	$2.33(8)\cdot 10^{3}$
0.255	$2.1634(3)\cdot 10^{0}$	$8.21(1)\cdot 10^{1}$	$2.35(7)\cdot 10^{3}$
0.265	$1.9062(3)\cdot 10^{0}$	$7.54(1)\cdot 10^{1}$	$2.15(7)\cdot 10^{3}$
0.275	$1.6470(3)\cdot 10^{0}$	$6.94(1)\cdot 10^{1}$	$1.97(6)\cdot 10^{3}$
0.285	$1.3841(3)\cdot 10^{0}$	$6.30(1)\cdot 10^{1}$	$1.80(6)\cdot 10^{3}$
0.295	$1.1169(2)\cdot 10^{0}$	$5.64(1)\cdot 10^{1}$	$1.83(5)\cdot 10^{3}$
0.305	$8.415(2)\cdot 10^{-1}$	$4.888(9)\cdot 10^{1}$	$1.74(5)\cdot 10^{3}$
0.315	$5.567(2)\cdot 10^{-1}$	$4.135(8)\cdot 10^{1}$	$1.57(4)\cdot 10^{3}$
0.325	$2.593(1)\cdot 10^{-1}$	$3.353(5)\cdot 10^{1}$	$1.42(2)\cdot 10^{3}$
0.335	$1.706(3)\cdot 10^{-2}$	$2.108(3)\cdot 10^{1}$	$1.03(1)\cdot 10^{3}$
0.345	0	$8.570(8)\cdot 10^{0}$	$2.96(5)\cdot 10^{2}$
0.355	0	$4.474(5)\cdot 10^{0}$	$1.55(3)\cdot 10^{2}$
0.365	0	$2.434(3)\cdot 10^{0}$	$5.3(1)\cdot 10^{1}$
0.375	0	$1.301(2)\cdot 10^{0}$	$3.57(7)\cdot 10^{1}$
0.385	0	$6.44(1)\cdot 10^{-1}$	$1.09(2)\cdot 10^{1}$
0.395	0	$2.818(8)\cdot 10^{-1}$	$1.8(2)\cdot 10^{-1}$
0.405	0	$5.12(3)\cdot 10^{-2}$	$-1.0(1)\cdot 10^{-3}$
0.415	0	$1.45987(8)\cdot 10^{-5}$	$-1.950(2)\cdot 10^{-1}$
0.425	0	$4.66279(3)\cdot 10^{-5}$	$-9(4)\cdot 10^{-3}$
0.435	0	0	$-1(1)\cdot 10^{-3}$
0.445	0	0	0

Table 1: Coefficients of the leading-order ($A_{(1-T)}$), next-to-leading-order ($B_{(1-T)}$) and next-to-next-to-leading order ($C_{(1-T)}$) contributions to the thrust distribution.

C	$C\frac{dA_C}{dC}$	$C\frac{dB_C}{dC}$	$C\frac{dC_C}{dC}$
0.005	$3.5327(6)\cdot 10^{1}$	$-2.151(1)\cdot 10^{3}$	$3.06(4)\cdot 10^{4}$
0.015	$2.7963(3)\cdot 10^{1}$	$-4.75(1)\cdot 10^{2}$	$-3.04(4)\cdot 10^{4}$
0.025	$2.5124(3)\cdot 10^{1}$	$-1.45(1)\cdot 10^{2}$	$-2.15(4)\cdot 10^{4}$
0.035	$2.3269(2)\cdot 10^{1}$	$1.3(1)\cdot 10^{1}$	$-1.48(4)\cdot 10^{4}$
0.045	$2.1887(2)\cdot 10^{1}$	$1.09(1)\cdot 10^{2}$	$-1.12(5)\cdot 10^{4}$
0.055	$2.0774(2)\cdot 10^{1}$	$1.71(1)\cdot 10^{2}$	$-7.6(5)\cdot 10^{3}$
0.065	$1.9831(2)\cdot 10^{1}$	$2.15(1)\cdot 10^{2}$	$-5.3(5)\cdot 10^{3}$
0.075	$1.9042(2)\cdot 10^{1}$	$2.44(1)\cdot 10^{2}$	$-2.7(5)\cdot 10^{3}$
0.085	$1.8340(2)\cdot 10^{1}$	$2.67(1)\cdot 10^{2}$	$-1.5(5)\cdot 10^{3}$
0.095	$1.7710(2)\cdot 10^{1}$	$2.83(1)\cdot 10^{2}$	$-7(5)\cdot 10^{2}$
0.105	$1.7152(2)\cdot 10^{1}$	$2.95(1)\cdot 10^{2}$	$1.6(5)\cdot 10^{3}$
0.115	$1.6637(2)\cdot 10^{1}$	$3.06(1)\cdot 10^{2}$	$1.4(5)\cdot 10^{3}$
0.125	$1.6154(2)\cdot 10^{1}$	$3.10(1)\cdot 10^{2}$	$1.7(5)\cdot 10^{3}$
0.135	$1.5713(2)\cdot 10^{1}$	$3.18(1)\cdot 10^{2}$	$3.3(5)\cdot 10^{3}$
0.145	$1.5312(2)\cdot 10^{1}$	$3.21(1)\cdot 10^{2}$	$3.6(5)\cdot 10^{3}$
0.155	$1.4933(2)\cdot 10^{1}$	$3.23(1)\cdot 10^{2}$	$4.3(5)\cdot 10^{3}$
0.165	$1.4572(2)\cdot 10^{1}$	$3.24(1)\cdot 10^{2}$	$4.2(5)\cdot 10^{3}$
0.175	$1.4228(2)\cdot 10^{1}$	$3.26(1)\cdot 10^{2}$	$4.8(5)\cdot 10^{3}$
0.185	$1.3915(2)\cdot 10^{1}$	$3.241(9)\cdot 10^{2}$	$5.4(5)\cdot 10^{3}$
0.195	$1.3609(2)\cdot 10^{1}$	$3.227(9)\cdot 10^{2}$	$4.4(5)\cdot 10^{3}$
0.205	$1.3321(2)\cdot 10^{1}$	$3.242(9)\cdot 10^{2}$	$5.5(5)\cdot 10^{3}$
0.215	$1.3039(2)\cdot 10^{1}$	$3.214(9)\cdot 10^{2}$	$5.8(5)\cdot 10^{3}$
0.225	$1.2779(2)\cdot 10^{1}$	$3.208(9)\cdot 10^{2}$	$5.9(5)\cdot 10^{3}$
0.235	$1.2524(2)\cdot 10^{1}$	$3.184(9)\cdot 10^{2}$	$6.2(5)\cdot 10^{3}$
0.245	$1.2281(2)\cdot 10^{1}$	$3.179(9)\cdot 10^{2}$	$6.7(5)\cdot 10^{3}$
0.255	$1.2046(2)\cdot 10^{1}$	$3.156(9)\cdot 10^{2}$	$6.3(5)\cdot 10^{3}$
0.265	$1.1823(2)\cdot 10^{1}$	$3.113(9)\cdot 10^{2}$	$6.2(5)\cdot 10^{3}$
0.275	$1.1608(2)\cdot 10^{1}$	$3.079(9)\cdot 10^{2}$	$6.6(5)\cdot 10^{3}$
0.285	$1.1392(2)\cdot 10^{1}$	$3.078(9)\cdot 10^{2}$	$5.9(5)\cdot 10^{3}$
0.295	$1.1197(2)\cdot 10^{1}$	$3.029(8)\cdot 10^{2}$	$6.8(5)\cdot 10^{3}$
0.305	$1.1000(2)\cdot 10^{1}$	$3.008(8)\cdot 10^{2}$	$6.5(5)\cdot 10^{3}$
0.315	$1.0812(2)\cdot 10^{1}$	$2.980(8)\cdot 10^{2}$	$6.8(5)\cdot 10^{3}$
0.325	$1.0627(2)\cdot 10^{1}$	$2.964(8)\cdot 10^{2}$	$6.5(5)\cdot 10^{3}$
0.335	$1.0451(1)\cdot 10^{1}$	$2.908(8)\cdot 10^{2}$	$6.3(5)\cdot 10^{3}$
0.345	$1.0273(1)\cdot 10^{1}$	$2.897(8)\cdot 10^{2}$	$6.1(5)\cdot 10^{3}$
0.355	$1.0110(1)\cdot 10^{1}$	$2.856(8)\cdot 10^{2}$	$6.2(5)\cdot 10^{3}$
0.365	$9.946(1)\cdot 10^{0}$	$2.843(8)\cdot 10^{2}$	$6.4(5)\cdot 10^{3}$
0.375	$9.787(1)\cdot 10^{0}$	$2.804(8)\cdot 10^{2}$	$7.0(5)\cdot 10^{3}$
0.385	$9.628(1)\cdot 10^{0}$	$2.752(8)\cdot 10^{2}$	$6.5(5)\cdot 10^{3}$
0.395	$9.484(1)\cdot 10^{0}$	$2.732(8)\cdot 10^{2}$	$6.1(5)\cdot 10^{3}$
0.405	$9.332(1)\cdot 10^{0}$	$2.711(8)\cdot 10^{2}$	$6.2(5)\cdot 10^{3}$
0.415	$9.190(1)\cdot 10^{0}$	$2.685(7)\cdot 10^{2}$	$7.0(5)\cdot 10^{3}$
0.425	$9.049(1)\cdot 10^{0}$	$2.636(7)\cdot 10^{2}$	$5.8(4)\cdot 10^{3}$
0.435	$8.914(1)\cdot 10^{0}$	$2.616(7)\cdot 10^{2}$	$6.1(4)\cdot 10^{3}$
0.445	$8.779(1)\cdot 10^{0}$	$2.580(7)\cdot 10^{2}$	$6.2(4)\cdot 10^{3}$
0.455	$8.645(1)\cdot 10^{0}$	$2.555(7)\cdot 10^{2}$	$6.0(4)\cdot 10^{3}$
0.465	$8.518(1)\cdot 10^{0}$	$2.518(7)\cdot 10^{2}$	$6.5(4)\cdot 10^{3}$
0.475	$8.392(1)\cdot 10^{0}$	$2.491(7)\cdot 10^{2}$	$5.6(4)\cdot 10^{3}$
0.485	$8.269(1)\cdot 10^{0}$	$2.465(7)\cdot 10^{2}$	$5.9(4)\cdot 10^{3}$
0.495	$8.152(1)\cdot 10^{0}$	$2.415(7)\cdot 10^{2}$	$5.8(4)\cdot 10^{3}$

Table 5: Coefficients of the leading-order (A_C), next-to-leading-order (B_C) and next-to-next-to-leading order (C_C) contributions to the C-parameter distribution.

C	$C\frac{dA_C}{dC}$	$C\frac{dB_C}{dC}$	$C\frac{dC_C}{dC}$
0.505	$8.033(1)\cdot 10^{0}$	$2.417(7)\cdot 10^{2}$	$5.8(4)\cdot 10^{3}$
0.515	$7.918(1)\cdot 10^{0}$	$2.376(7)\cdot 10^{2}$	$5.5(4)\cdot 10^{3}$
0.525	$7.804(1)\cdot 10^{0}$	$2.326(7)\cdot 10^{2}$	$6.1(4)\cdot 10^{3}$
0.535	$7.691(1)\cdot 10^{0}$	$2.312(7)\cdot 10^{2}$	$5.0(4)\cdot 10^{3}$
0.545	$7.589(1)\cdot 10^{0}$	$2.285(7)\cdot 10^{2}$	$5.5(4)\cdot 10^{3}$
0.555	$7.476(1)\cdot 10^{0}$	$2.247(7)\cdot 10^{2}$	$5.7(4)\cdot 10^{3}$
0.565	$7.377(1)\cdot 10^{0}$	$2.227(7)\cdot 10^{2}$	$5.7(4)\cdot 10^{3}$
0.575	$7.271(1)\cdot 10^{0}$	$2.190(7)\cdot 10^{2}$	$5.6(4)\cdot 10^{3}$
0.585	$7.172(1)\cdot 10^{0}$	$2.180(7)\cdot 10^{2}$	$5.4(4)\cdot 10^{3}$
0.595	$7.073(1)\cdot 10^{0}$	$2.143(7)\cdot 10^{2}$	$5.2(4)\cdot 10^{3}$
0.605	$6.976(1)\cdot 10^{0}$	$2.105(7)\cdot 10^{2}$	$4.9(4)\cdot 10^{3}$
0.615	$6.880(1)\cdot 10^{0}$	$2.084(7)\cdot 10^{2}$	$5.2(4)\cdot 10^{3}$
0.625	$6.788(1)\cdot 10^{0}$	$2.046(7)\cdot 10^{2}$	$4.6(4)\cdot 10^{3}$
0.635	$6.692(1)\cdot 10^{0}$	$2.025(7)\cdot 10^{2}$	$5.1(4)\cdot 10^{3}$
0.645	$6.600(1)\cdot 10^{0}$	$2.009(7)\cdot 10^{2}$	$5.1(4)\cdot 10^{3}$
0.655	$6.515(1)\cdot 10^{0}$	$1.977(7)\cdot 10^{2}$	$4.4(4)\cdot 10^{3}$
0.665	$6.426(1)\cdot 10^{0}$	$1.947(7)\cdot 10^{2}$	$5.0(4)\cdot 10^{3}$
0.675	$6.337(1)\cdot 10^{0}$	$1.924(7)\cdot 10^{2}$	$4.7(4)\cdot 10^{3}$
0.685	$6.253(1)\cdot 10^{0}$	$1.892(7)\cdot 10^{2}$	$5.1(4)\cdot 10^{3}$
0.695	$6.171(1)\cdot 10^{0}$	$1.877(7)\cdot 10^{2}$	$4.5(4)\cdot 10^{3}$
0.705	$6.0874(9)\cdot 10^{0}$	$1.840(7)\cdot 10^{2}$	$4.3(3)\cdot 10^{3}$
0.715	$6.0103(9)\cdot 10^{0}$	$1.826(7)\cdot 10^{2}$	$3.9(3)\cdot 10^{3}$
0.725	$5.9266(9)\cdot 10^{0}$	$1.800(7)\cdot 10^{2}$	$4.4(3)\cdot 10^{3}$
0.735	$5.8474(9)\cdot 10^{0}$	$1.775(7)\cdot 10^{2}$	$4.3(3)\cdot 10^{3}$
0.745	$5.7695(9)\cdot 10^{0}$	$1.745(8)\cdot 10^{2}$	$3.2(3)\cdot 10^{3}$
0.755	0	$6.011(5)\cdot 10^{2}$	$1.27(2)\cdot 10^{4}$
0.765	0	$2.771(2)\cdot 10^{2}$	$1.38(1)\cdot 10^{4}$
0.775	0	$1.905(1)\cdot 10^{2}$	$9.99(9)\cdot 10^{3}$
0.785	0	$1.4246(7)\cdot 10^{2}$	$7.66(7)\cdot 10^{3}$
0.795	0	$1.1124(6)\cdot 10^{2}$	$6.17(6)\cdot 10^{3}$
0.805	0	$8.895(5)\cdot 10^{1}$	$4.65(5)\cdot 10^{3}$
0.815	0	$7.228(4)\cdot 10^{1}$	$3.49(4)\cdot 10^{3}$
0.825	0	$5.913(3)\cdot 10^{1}$	$3.02(4)\cdot 10^{3}$
0.835	0	$4.876(3)\cdot 10^{1}$	$2.50(3)\cdot 10^{3}$
0.845	0	$4.037(2)\cdot 10^{1}$	$1.93(3)\cdot 10^{3}$
0.855	0	$3.336(2)\cdot 10^{1}$	$8.2(2)\cdot 10^{2}$
0.865	0	$2.754(2)\cdot 10^{1}$	$1.26(2)\cdot 10^{3}$
0.875	0	$2.270(2)\cdot 10^{1}$	$9.4(2)\cdot 10^{2}$
0.885	0	$1.856(1)\cdot 10^{1}$	$6.8(1)\cdot 10^{2}$
0.895	0	$1.506(1)\cdot 10^{1}$	$5.9(1)\cdot 10^{2}$
0.905	0	$1.212(1)\cdot 10^{1}$	$4.57(9)\cdot 10^{2}$
0.915	0	$9.587(9)\cdot 10^{0}$	$3.50(7)\cdot 10^{2}$
0.925	0	$7.458(8)\cdot 10^{0}$	$2.19(5)\cdot 10^{2}$
0.935	0	$5.645(7)\cdot 10^{0}$	$1.0(2)\cdot 10^{1}$
0.945	0	$4.125(6)\cdot 10^{0}$	$1.01(2)\cdot 10^{2}$
0.955	0	$2.875(5)\cdot 10^{0}$	$2.59(8)\cdot 10^{1}$
0.965	0	$1.835(4)\cdot 10^{0}$	$2.26(5)\cdot 10^{1}$
0.975	0	$1.008(3)\cdot 10^{0}$	$3.9(1)\cdot 10^{0}$
0.985	0	$3.99(1)\cdot 10^{-1}$	$-4.886(1)\cdot 10^{-1}$
0.995	0	$1.8841(2)\cdot 10^{-2}$	$-1.443(2)\cdot 10^{-1}$

Table 6: Coefficients of the leading-order (A_C), next-to-leading-order (B_C) and next-to-next-to-leading order (C_C) contributions to the C-parameter distribution.

Unbarred coefficients from Weinzierl *

C++ structure for coefficients of Thrust and C-parameter

```cpp
// Thrust [Bins, LO, NLO, NNLO]
std::vector<std::array<double,4>> Theory_THR = {
    {0.005, 25.7970, -495.7000, -9000.0000},
    {0.015, 18.2990,  216.6000, -3300.0000},
    {0.025, 15.4110,  280.9000,  1600.0000},
    {0.035, 13.5230,  292.0000,  4000.0000},
    {0.045, 12.1030,  287.7000,  4900.0000},
    {0.055, 10.9692,  277.2000,  5300.0000},
    {0.065, 10.0228,  263.8000,  5500.0000},
    {0.075,  9.2112,  250.5000,  5400.0000},
    {0.085,  8.5022,  237.1000,  5500.0000},
    {0.095,  7.8724,  224.4000,  5200.0000},
    {0.105,  7.3045,  211.3000,  5100.0000},
    {0.115,  6.7891,  199.3000,  4800.0000},
    {0.125,  6.3179,  187.8000,  4500.0000},
    {0.135,  5.8820,  176.8000,  4300.0000},
    {0.145,  5.4791,  166.6000,  4100.0000},
    {0.155,  5.1030,  157.2000,  3900.0000},
    {0.165,  4.7484,  147.5000,  3800.0000},
    {0.175,  4.4141,  139.3000,  3500.0000},
    {0.185,  4.0962,  130.6000,  3290.0000},
    {0.195,  3.7931,  123.1000,  3200.0000},
    {0.205,  3.5028,  115.3000,  2980.0000},
    {0.215,  3.2220,  108.1000,  2850.0000},
    {0.225,  2.9494,  101.1000,  2630.0000},
    {0.235,  2.6826,   94.7000,  2500.0000},
    {0.245,  2.4223,   88.2000,  2330.0000},
    {0.255,  2.1634,   82.1000,  2350.0000},
    {0.265,  1.9062,   75.4000,  2150.0000},
    {0.275,  1.6470,   69.4000,  1970.0000},
    {0.285,  1.3841,   63.0000,  1800.0000},
    {0.295,  1.1169,   56.4000,  1830.0000},
    {0.305,  0.8415,   48.8800,  1740.0000},
    {0.315,  0.5567,   41.3500,  1570.0000},
    {0.325,  0.2593,   33.5300,  1420.0000},
    {0.335,  0.1706,   21.0800,  1030.0000},
    {0.345,  0.0000,    8.5700,   296.0000},
    {0.355,  0.0000,    4.4740,   155.0000},
    {0.365,  0.0000,    2.4340,    53.0000},
    {0.375,  0.0000,    1.3010,    35.7000},
    {0.385,  0.0000,    0.6440,    10.9000},
    {0.395,  0.0000,    0.2818,     0.1800},
    {0.405,  0.0000,    5.12E-2,   -1.0E-3},
    {0.415,  0.0000,    1.45987E-5,-0.1950},
    {0.425,  0.0000,    4.66279E-5,-0.0090},
    {0.435,  0.0000,    0.0000,    -0.0010},
    {0.445,  0.0000,    0.0000,     0.0000}
};
```

```cpp
// C-parameter [Bins, LO, NLO, NNLO]
std::vector<std::array<double,4>> Theory_CPR = {
    {0.005, 35.3270, -2151.0000, 30600.0000}, {0.015, 27.9630,  -475.1000, -30400.0000},
    {0.025, 25.1240,  -145.1000, -21500.0000}, {0.035, 23.2690,    13.0000, -14800.0000},
    {0.045, 21.8870,   109.1000, -11200.0000}, {0.055, 20.7740,   171.0000,  -7600.0000},
    {0.065, 19.8310,   215.0000,  -5300.0000}, {0.075, 19.0420,   244.0000,  -2700.0000},
    {0.085, 18.3400,   267.0000,  -1500.0000}, {0.095, 17.7100,   283.0000,   -750.0000},
    {0.105, 17.1520,   295.0000,   1600.0000}, {0.115, 16.6370,   306.0000,   1400.0000},
    {0.125, 16.1540,   310.0000,   1700.0000}, {0.135, 15.7130,   318.0000,   3300.0000},
    {0.145, 15.3120,   321.0000,   3600.0000}, {0.155, 14.9330,   323.0000,   4300.0000},
    {0.165, 14.5720,   324.0000,   4200.0000}, {0.175, 14.2280,   326.0000,   4800.0000},
    {0.185, 13.9150,   324.1000,   5400.0000}, {0.195, 13.6090,   322.7000,   4400.0000},
    {0.205, 13.3210,   324.2000,   5500.0000}, {0.215, 13.0390,   321.4000,   5800.0000},
    {0.225, 12.7790,   320.8000,   5900.0000}, {0.235, 12.5240,   318.4000,   6200.0000},
    {0.245, 12.2810,   317.9000,   6700.0000}, {0.255, 12.0460,   315.6000,   6300.0000},
    {0.265, 11.8230,   311.3000,   6200.0000}, {0.275, 11.6080,   307.9000,   6600.0000},
    {0.285, 11.3920,   307.8000,   5900.0000}, {0.295, 11.1970,   302.9000,   6800.0000},
    {0.305, 11.0000,   300.8000,   6500.0000}, {0.315, 10.8120,   298.0000,   6800.0000},
    {0.325, 10.6270,   296.4000,   6500.0000}, {0.335, 10.4510,   290.8000,   6300.0000},
    {0.345, 10.2730,   289.7000,   6100.0000}, {0.355, 10.1100,   285.6000,   6200.0000},
    {0.365,  9.9460,   284.3000,   6400.0000}, {0.375,  9.7870,   280.4000,   7000.0000},
    {0.385,  9.6280,   275.2000,   6500.0000}, {0.395,  9.4840,   273.2000,   6100.0000},
    {0.405,  9.3321,   271.1000,   6200.0000}, {0.415,  9.1900,   268.5000,   7000.0000},
    {0.425,  9.0490,   263.6000,   5800.0000}, {0.435,  8.9140,   261.6000,   6100.0000},
    {0.445,  8.7791,   258.0000,   6200.0000}, {0.455,  8.6450,   255.5000,   6000.0000},
    {0.465,  8.5180,   251.8000,   6500.0000}, {0.475,  8.3920,   249.1000,   5600.0000},
    {0.485,  8.2690,   246.5000,   5900.0000}, {0.495,  8.1520,   241.5000,   5800.0000},
    {0.505,  8.0330,   241.7000,   5800.0000}, {0.515,  7.9180,   237.6000,   5500.0000},
    {0.525,  7.8040,   232.6000,   6100.0000}, {0.535,  7.6910,   231.2000,   5000.0000},
    {0.545,  7.5890,   228.5000,   5500.0000}, {0.555,  7.4760,   224.7000,   5700.0000},
    {0.565,  7.3770,   224.7000,   5700.0000}, {0.575,  7.2710,   219.0000,   5600.0000},
    {0.585,  7.1720,   218.0000,   5400.0000}, {0.595,  7.0730,   214.3000,   5200.0000},
    {0.605,  6.9760,   210.5000,   4900.0000}, {0.615,  6.8800,   208.4000,   5200.0000},
    {0.625,  6.7880,   204.6000,   4600.0000}, {0.635,  6.6920,   202.5000,   5100.0000},
    {0.645,  6.6000,   200.9000,   5100.0000}, {0.655,  6.5150,   197.7000,   4400.0000},
    {0.665,  6.4260,   194.7000,   5000.0000}, {0.675,  6.3370,   192.4000,   4700.0000},
    {0.685,  6.2530,   189.2000,   5100.0000}, {0.695,  6.1710,   187.7000,   4500.0000},
    {0.705,  6.0874,   184.0000,   4300.0000}, {0.715,  6.0103,   182.6000,   3900.0000},
    {0.725,  5.9266,   180.0000,   4400.0000}, {0.735,  5.8474,   177.5000,   4300.0000},
    {0.745,  5.7695,   174.5000,   3200.0000}, {0.755,  0.0000,   601.1000,  12700.0000},
    {0.765,  0.0000,   277.1000,  13800.0000}, {0.775,  0.0000,   190.5000,   9990.0000},
    {0.785,  0.0000,   142.4600,   7600.0000}, {0.795,  0.0000,   111.2400,   6170.0000},
    {0.805,  0.0000,    88.9500,   4650.0000}, {0.815,  0.0000,    72.2800,   3490.0000},
    {0.825,  0.0000,    59.1300,   3020.0000}, {0.835,  0.0000,    48.7600,   1930.0000},
    {0.845,  0.0000,    40.3700,   1930.0000}, {0.855,  0.0000,    33.3600,    820.0000},
    {0.865,  0.0000,    27.5400,   1260.0000}, {0.875,  0.0000,    22.7000,    940.0000},
    {0.885,  0.0000,    18.5600,    680.0000}, {0.895,  0.0000,    15.0600,    590.0000},
    {0.905,  0.0000,    12.1200,    457.0000}, {0.915,  0.0000,     9.5870,    350.0000},
    {0.925,  0.0000,     7.4580,    219.0000}, {0.935,  0.0000,     5.6450,     10.0000},
    {0.945,  0.0000,     4.1250,    101.9000}, {0.955,  0.0000,     2.8750,     25.9000},
    {0.965,  0.0000,     1.8350,     22.6000}, {0.975,  0.0000,     1.0080,      3.9000},
    {0.985,  0.0000,     0.3990,     -0.4886}, {0.995,  0.0000,     0.0188,     -0.1443}
};
```

* S. Weinzierl, Event shapes and jet rates in electron-positron annihilation at NNLO, Journal of High Energy Physics 2009, 041–041 (2009)

Workflow – Theory functions

Perturbative expansion with barred coefficients

The perturbative expansion of a differential distribution for any infrared-safe observable O for the process $e^+e^- \to 3$ jets can be written up to NNLO as

$$\frac{O^n}{\sigma_{tot}(\mu)}\frac{d\sigma(\mu)}{dO} = \frac{\alpha_s(\mu)}{2\pi}\frac{O^n d\bar{A}_O(\mu)}{dO} + \left(\frac{\alpha_s(\mu)}{2\pi}\right)^2 \frac{O^n d\bar{B}_O(\mu)}{dO} + \left(\frac{\alpha_s(\mu)}{2\pi}\right)^3 \frac{O^n d\bar{C}_O(\mu)}{dO}. \quad (17)$$

$\bar{A}_O$ gives the LO result, $\bar{B}_O$ the NLO correction and $\bar{C}_O$ the NNLO correction. The variable n denotes the moment of the distribution. Unless stated otherwise, the value $n = 1$ is used as default. σ_{tot} denotes the total hadronic cross section calculated up to the relevant order. The arbitrary renormalisation scale is denoted by μ. The n-th moment is given by

$$\langle O^n \rangle = \frac{1}{\sigma_{tot}} \int O^n \frac{d\sigma}{dO} dO. \quad (18)$$

In practise the numerical program computes the distribution

$$\boxed{\frac{O^n}{\sigma_0(\mu)}\frac{d\sigma(\mu)}{dO} = \frac{\alpha_s(\mu)}{2\pi}\frac{O^n dA_O(\mu)}{dO} + \left(\frac{\alpha_s(\mu)}{2\pi}\right)^2 \frac{O^n dB_O(\mu)}{dO} + \left(\frac{\alpha_s(\mu)}{2\pi}\right)^3 \frac{O^n dC_O(\mu)}{dO}} \quad (19)$$

normalised to σ_0, which is the LO cross section for $e^+e^- \to$ hadrons, instead of the normalisation to σ_{tot}. There is a simple relation between the two distributions: The functions A_O, B_O and C_O are related to the functions $\bar{A}_O$, $\bar{B}_O$ and $\bar{C}_O$ by

$$\boxed{\begin{aligned} \bar{A}_O &= A_O, \\ \bar{B}_O &= B_O - A_{tot}A_O, \\ \bar{C}_O &= C_O - A_{tot}B_O - (B_{tot} - A_{tot}^2)A_O, \end{aligned}}$$

where

$$\begin{aligned} A_{tot} &= \frac{3(N_c^2-1)}{4N_c}, \\ B_{tot} &= \frac{N_c^2-1}{8N_c}\left[\left(\frac{243}{4} - 44\zeta_3\right)N_c + \frac{3}{4N_c} + (8\zeta_3 - 11)N_f\right]. \end{aligned}$$

Theory function for (1–T) and C-parameter

```
double THR_NNLO(double *x, double *par) {

    // fit bin call
    double tau = x[0];
    // fit variable
    double alphaS = par[0];

    // number of bins
    const int nBins = Theory_THR.size();
    // first theory bin
    const double firstBin = Theory_THR[0][0];
    // second theory bin
    const double secndBin = Theory_THR[1][0];
    // bin width
    const double step = secndBin - firstBin;

    // find nearest bin
    int idx = int( (tau - firstBin)/step + 0.5 );
    if (idx < 0) idx = 0;
    if (idx >= nBins) idx = nBins - 1;

    // selected bin index
    const auto &row = Theory_THR[idx];
    double bin  = row[0];
    double A    = row[1];
    double B    = row[2];
    double C    = row[3];

    // barred
    double AA = A;
    double BB = B - At*A;
    double CC = C - At*B - (Bt - At*At)*A;

    // construct
    double asbar = alphaS / (2 * TMath::Pi());
    double val = asbar*AA + asbar*asbar*BB + asbar*asbar*asbar*CC;

    // normalise
    return val / bin;
}
```

= divide by bin width

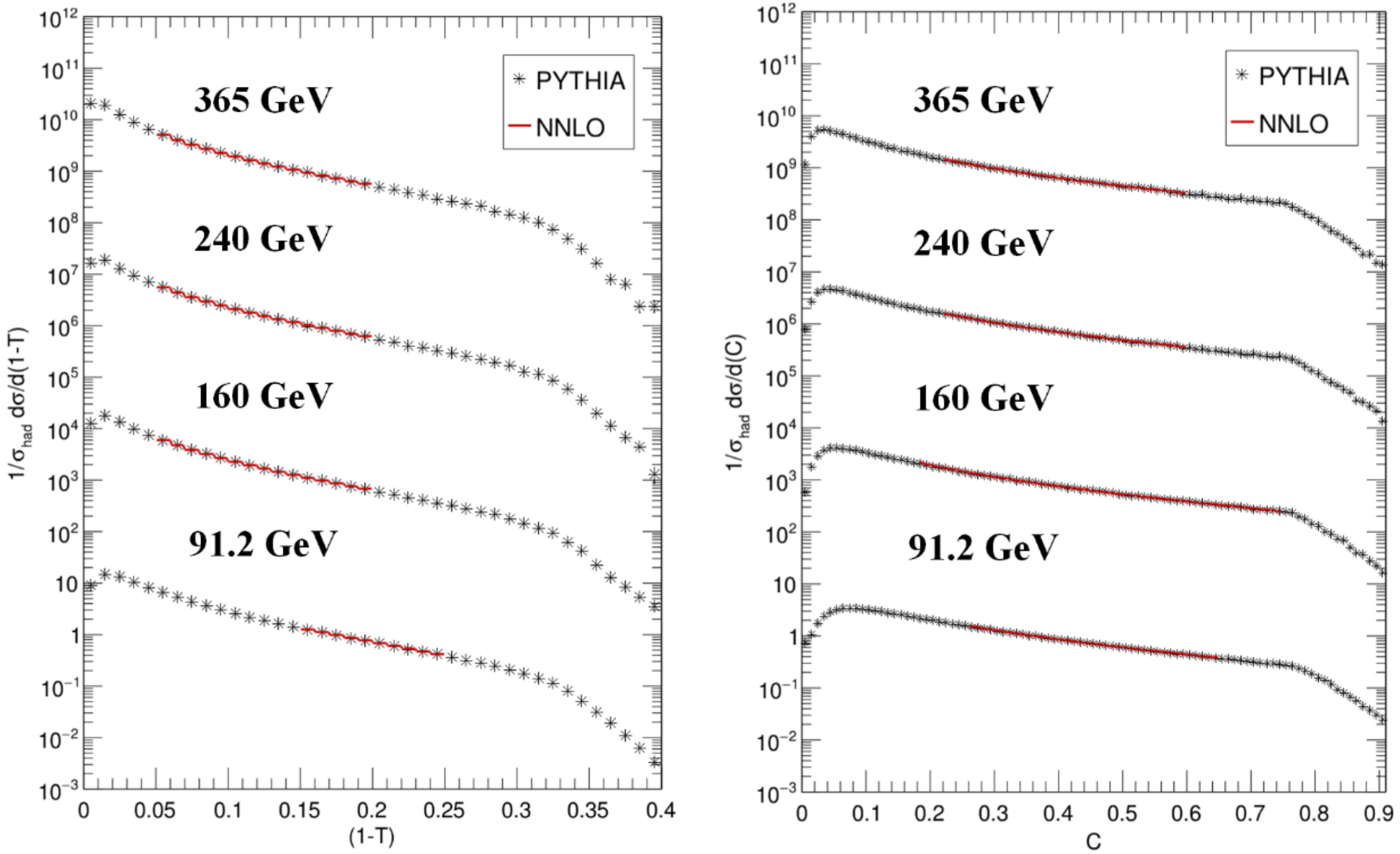
```
// unbarred conversions
double At = 3.0*(Nc*Nc-1.0)/(4.0*Nc);
double Bt = (Nc*Nc-1.0)/(8.0*Nc) * ( (243.0/4.0 - 44.0*1.202)*Nc + 3.0/(4.0*Nc) + (8.0*1.202 - 11.0)*Nf );
```

Workflow – TF1 functions

```
TF1 *fits_fitThNN_912 = new TF1("fits_fitThNN_912", THR_NNLO, 0.15, 0.25, 1);
fits_fitThNN_912->SetLineColor(kBlue+1); fits_fitThNN_912->SetMarkerColor(kBlue+1); fits_fitThNN_912->SetMarkerStyle(53); fits_fitThNN_912->SetLineWidth(2); fits_fitThNN_912->SetMarkerSize(1);
TF1 *fits_fitThNN_160 = new TF1("fits_fitThNN_160", THR_NNLO, 0.08, 0.20, 1);
fits_fitThNN_160->SetLineColor(kBlue+1); fits_fitThNN_160->SetMarkerColor(kBlue+1); fits_fitThNN_160->SetMarkerStyle(53); fits_fitThNN_160->SetLineWidth(2); fits_fitThNN_160->SetMarkerSize(1);
TF1 *fits_fitThNN_240 = new TF1("fits_fitThNN_240", THR_NNLO, 0.05, 0.20, 1);
fits_fitThNN_240->SetLineColor(kBlue+1); fits_fitThNN_240->SetMarkerColor(kBlue+1); fits_fitThNN_240->SetMarkerStyle(53); fits_fitThNN_240->SetLineWidth(2); fits_fitThNN_240->SetMarkerSize(1);
TF1 *fits_fitThNN_365 = new TF1("fits_fitThNN_365", THR_NNLO, 0.05, 0.20, 1);
fits_fitThNN_365->SetLineColor(kBlue+1); fits_fitThNN_365->SetMarkerColor(kBlue+1); fits_fitThNN_365->SetMarkerStyle(53); fits_fitThNN_365->SetLineWidth(2); fits_fitThNN_365->SetMarkerSize(1);

TF1 *fits_fitCpNN_912 = new TF1("fits_fitCpNN_912", CPR_NNLO, 0.45, 0.60, 1);
fits_fitCpNN_912->SetLineColor(kBlue+1); fits_fitCpNN_912->SetMarkerColor(kBlue+1); fits_fitCpNN_912->SetMarkerStyle(53); fits_fitCpNN_912->SetLineWidth(2); fits_fitCpNN_912->SetMarkerSize(1);
TF1 *fits_fitCpNN_160 = new TF1("fits_fitCpNN_160", CPR_NNLO, 0.25, 0.75, 1);
fits_fitCpNN_160->SetLineColor(kBlue+1); fits_fitCpNN_160->SetMarkerColor(kBlue+1); fits_fitCpNN_160->SetMarkerStyle(53); fits_fitCpNN_160->SetLineWidth(2); fits_fitCpNN_160->SetMarkerSize(1);
TF1 *fits_fitCpNN_240 = new TF1("fits_fitCpNN_240", CPR_NNLO, 0.22, 0.60, 1);
fits_fitCpNN_240->SetLineColor(kBlue+1); fits_fitCpNN_240->SetMarkerColor(kBlue+1); fits_fitCpNN_240->SetMarkerStyle(53); fits_fitCpNN_240->SetLineWidth(2); fits_fitCpNN_240->SetMarkerSize(1);
TF1 *fits_fitCpNN_365 = new TF1("fits_fitCpNN_365", CPR_NNLO, 0.22, 0.60, 1);
fits_fitCpNN_365->SetLineColor(kBlue+1); fits_fitCpNN_365->SetMarkerColor(kBlue+1); fits_fitCpNN_365->SetMarkerStyle(53); fits_fitCpNN_365->SetLineWidth(2); fits_fitCpNN_365->SetMarkerSize(1);

TF1 *fits_fitThNN_91X = new TF1("fits_fitThNN_91X", THR_NNLO, 0.15, 0.25, 1);
fits_fitThNN_91X->SetLineColor(kRed+2); fits_fitThNN_91X->SetMarkerColor(kRed+2); fits_fitThNN_91X->SetMarkerStyle(53); fits_fitThNN_91X->SetLineWidth(2); fits_fitThNN_91X->SetMarkerSize(1);
TF1 *fits_fitCpNN_91X = new TF1("fits_fitCpNN_91X", CPR_NNLO, 0.65, 0.74, 1);
fits_fitCpNN_91X->SetLineColor(kRed+2); fits_fitCpNN_91X->SetMarkerColor(kRed+2); fits_fitCpNN_91X->SetMarkerStyle(53); fits_fitCpNN_91X->SetLineWidth(2); fits_fitCpNN_91X->SetMarkerSize(1);
```

```
hist_ThrPyth_912_woHadron->Fit(fits_fitThNN_912, "RNQ MINOS");
hist_ThrPyth_160_woHadron->Fit(fits_fitThNN_160, "RNQ MINOS");
hist_ThrPyth_240_woHadron->Fit(fits_fitThNN_240, "RNQ MINOS");
hist_ThrPyth_365_woHadron->Fit(fits_fitThNN_365, "RNQ MINOS");

hist_CprPyth_912_woHadron->Fit(fits_fitCpNN_912, "RNQ MINOS");
hist_CprPyth_160_woHadron->Fit(fits_fitCpNN_160, "RNQ MINOS");
hist_CprPyth_240_woHadron->Fit(fits_fitCpNN_240, "RNQ MINOS");
hist_CprPyth_365_woHadron->Fit(fits_fitCpNN_365, "RNQ MINOS");

hist_ThrALPH_912_wiHadron->Fit(fits_fitThNN_91X, "RNQ MINOS");
hist_CprALPH_912_wiHadron->Fit(fits_fitCpNN_91X, "RNQ MINOS");
```

Results – TF1 fits



Results – Extracted α_s

TABLE II. Strong coupling constant α_S obtained from fitting NNLO theory to PYTHIA event shapes at FCC-ee energies.

Shape	Result	91.2 GeV	160 GeV	240 GeV	365 GeV
$(1-T)$	α_S	0.1284	0.1190	0.1127	0.1064
	χ^2/N	9.3/9	18.8/11	24.7/14	14.4/14
	Range	0.15–0.25	0.08–0.20	0.05–0.20	0.05–0.20
	Error	±0.0001	±0.0002	±0.0002	±0.0002
C	α_S	0.1284	0.1183	0.1120	0.1058
	χ^2/N	25.9/14	84.8/49	65.6/37	58.8/37
	Range	0.45–0.60	0.25–0.75	0.22–0.60	0.22–0.60
	Error	±0.0001	±0.0001	±0.0002	±0.0002
Average	α_S	0.1284	0.1186	0.1123	0.1061
	Error	±0.0001	±0.0001	±0.0002	±0.0002

(Dissertori*) Contributions of event shapes to NNLO value

	T	C	M_H	B_W	B_T	$-\ln y_3$
NNLO+NLLA	0.1266	0.1252	0.1211	0.1196	0.1268	0.1186
χ^2/N_{dof}	0.16	0.47	4.4	4.4	0.84	1.89
stat.error	0.0002	0.0002	0.0003	0.0002	0.0002	0.0002
NLO+NLLA	0.1282	0.1244	0.1180	0.1161	0.1290	0.1187
χ^2/N_{dof}	0.74	1.88	14.5	19.6	9.7	4.7
stat.error	0.0002	0.0002	0.0003	0.0002	0.0002	0.0002
NNLO	0.1275	0.1273	0.1248	0.1242	0.1279	0.1192
χ^2/N_{dof}	1.16	1.08	4.1	2.74	0.50	1.17
stat.error	0.0002	0.0002	0.0004	0.0002	0.0002	0.0003
fit range	0.75 - 0.91	0.36 - 0.74	0.10 - 0.22	0.09 - 0.19	0.16 - 0.30	1.6-4.0

Table 3: Fit results for $\alpha_s(M_Z)$ using different predictions of perturbative QCD, with the renormalisation scale fixed to $\mu = M_Z$.

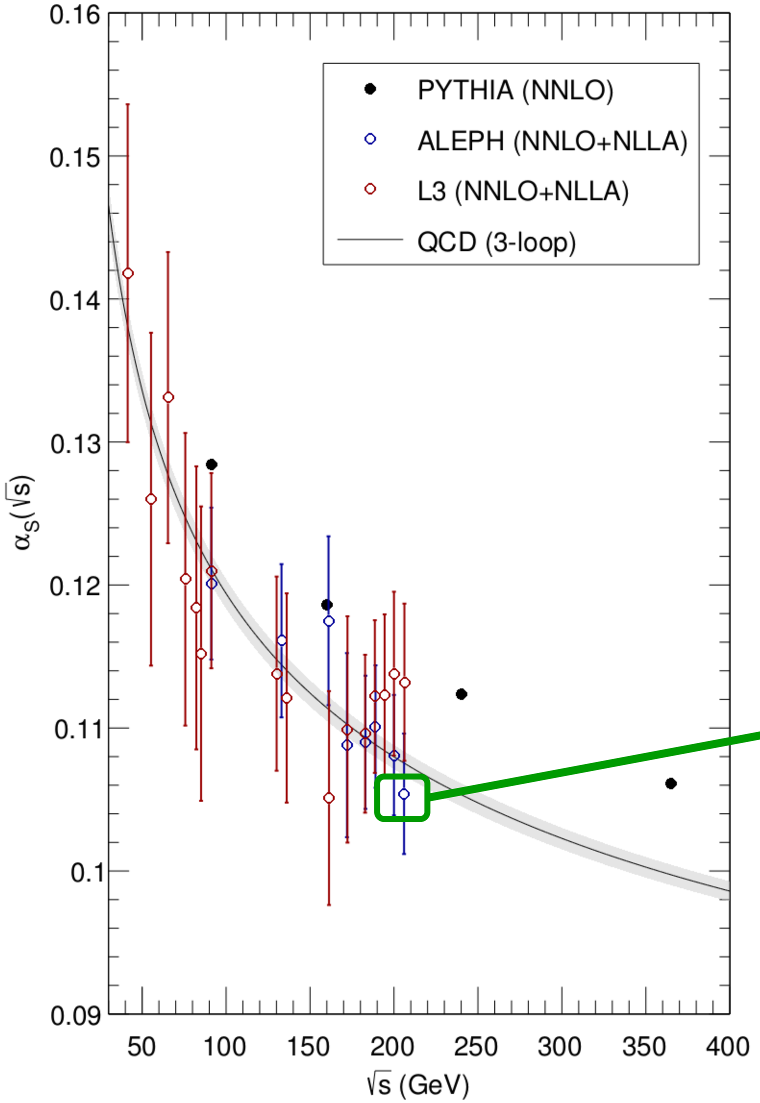
(Dissertori*) Extracted NNLO values at LEP-II energies

Q [GeV]	91.2	133	161	172	183	189	200	206
$\alpha_s(M_Z)$	0.1239	0.1270	0.1313	0.1192	0.1226	0.1234	0.1200	0.1202
stat. error	0.0002	0.0033	0.0051	0.0063	0.0028	0.0020	0.0021	0.0019
exp. error	0.0009	0.0009	0.0009	0.0010	0.0010	0.0010	0.0010	0.0010
pert. error	0.0030	0.0030	0.0028	0.0028	0.0027	0.0026	0.0025	0.0025
hadr. error	0.0018	0.0014	0.0012	0.0012	0.0011	0.0010	0.0010	0.0010
total error	0.0037	0.0048	0.0060	0.0070	0.0041	0.0036	0.0036	0.0034
RMS	0.0036	0.0014	0.0043	0.0019	0.0027	0.0027	0.0034	0.0024

Table 6: Combined results for $\alpha_s(M_Z)$ using NNLO predictions.

* S. Weinzierl, Event shapes and jet rates in electron-positron annihilation at NNLO, Journal of High Energy Physics 2009, 041–041 (2009)

Results – Energy evolution



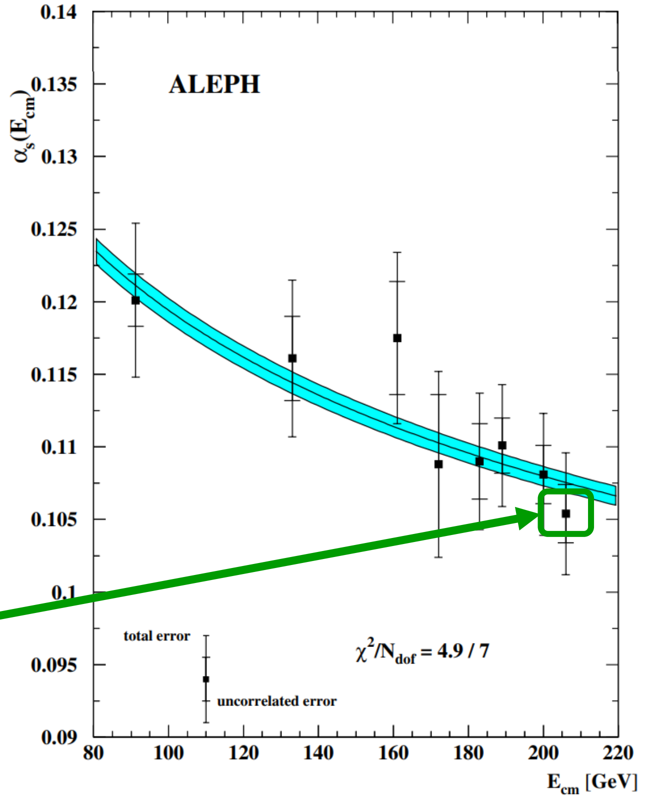


Fig. 14. The measurements of the strong coupling constant α_s between 91.2 and 206 GeV. The results using the six different event-shape variables are combined with correlations taken into account. The inner error bars exclude the perturbative uncertainty, which is expected to be highly correlated between the measurements. The outer error bars indicate the total error. A fit of the three-loop evolution formula using the uncorrelated errors is shown. The shaded area corresponds to the uncertainty in the fit parameter of the three-loop formula $\Lambda_{\overline{\mathrm{MS}}} = 247 \pm 14\,\mathrm{MeV}$, equivalent to $\alpha_s(M_Z) = 0.1211 \pm 0.0008$

* ALEPH, Studies of QCD at ee centre-of-mass energies between 91-GeV and 209-GeV, Eur. Phys. J. C 35, 457 (2004).

Summary of questions

- Regarding Topic 1:
  - Question 1.1
  - Question 1.2
- Regarding Topic 2:
  - Question 2.1
  - Question 2.2
- Regarding fit windows:
  - ALEPH ranges were used as baseline
  - Z-pole window needed to be shortened